EE 354 – SYNCHRONOUS MOTOR

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EE 354 Power Engineering Synchronous Motor

(3 hours)

AIM

To examine the characteristics of a synchronous motor.

APPARATUS

1. 3Φ Synchronous Motor (SM 100-3, 1425 rpm, Armature-230V AC / 1.6A, Field 220V DC / 0.6A)
2. Separately Excited DC Generator (DCG 100, 220V, 1800 rpm)
3. AC Voltmeter (0-750V) x 1
4. AC Ammeter (0-2A) x 1
5. DC Voltmeter (0-300V) x 1
6. DC Ammeter (0-3A) x 3
7. Rheostat (330Ω, 2A) x 1
8. Rheostat (500Ω, 1.6A) x 1
9. Wattmeter (240V, 5A) x 2
10. Bulb Load
11. Digital Tachometer

DATA

DC Generator;
Armature Winding Resistance, R_a =15Ω
Rotational Losses at 1500rpm, P_{rot}=30W

THEORY

Input power of the synchronous motor, $P_{in} = W_1 + W_2$

Copper loss of the DC generator, $P_{Cu} = I_A{}^2 \times R_A \leftarrow I_A = A_3$

Output power of the DC generator, $P_{DC} = V_T \times I_A \leftarrow I_A = A_3 \& V_T = V_2$

Output power of the synchronous motor, $P_{out} = P_{DC} + P_{cu} + P_{rot}$

Efficiency of the synchronous motor, $\eta = \frac{P_{out}}{P_{in}} \times 100\%$

PROCEDURE

1. Make sure DC supply (S2) is **switched off.**
2. Make sure AC supply (S1) is **switched off.**
3. Connect the circuit as shown in the **Figure 3**.
4. The synchronous motor should be star connected.
5. Keep both field rheostats (R1 and R2) at their **maximum resistive** positions.
6. Show the setup to the instructor.
7. Put DC field switch to the **SYNC-RUN** position.
8. Turn on **DC supply** (S2) of the **synchronous motor.**
9. Adjust the field rheostat (R1) of the synchronous motor until its field current becomes **0.4A** (A2 ammeter).
10. Return the switch to **IND-START** position.
11. **Supply AC** power to the stator (Switch on S1 switch).
12. Move the switch to **SYNC-RUN** position. Then, rotor will synchronize with the line and motor will be running at exactly synchronous speed (1500rpm).
13. Turn on **DC supply** of the **DC generator** to power up its field (switch on S3 switch).
14. Adjust DC generator field rheostat (R2) until its output voltage reaches **200V** (V_2 voltmeter).
15. While keeping the **synchronous motor excitation constant,** take the meter readings for **no load** position without any bulb load. Then, change the DC output power using different **combinations of bulb loads.** Combinations are **60W, 160W, 260W.**
16. Record the readings.
17. Repeat the step 15 and 16 for synchronous motor excitations from 0.4A to 1.2A in steps of 0.1A.

Shutdown Procedure

18. Increase the **DC generator rheostat (R2)** to its maximum and turn off all bulb loads.
19. Put the synchronous motor starting method to **IND-START** and turn off AC power supply (S1).
20. Increase the **synchronous motor rheostat (R1)** to its **maximum** position.
21. Turn off DC power supply (S2).

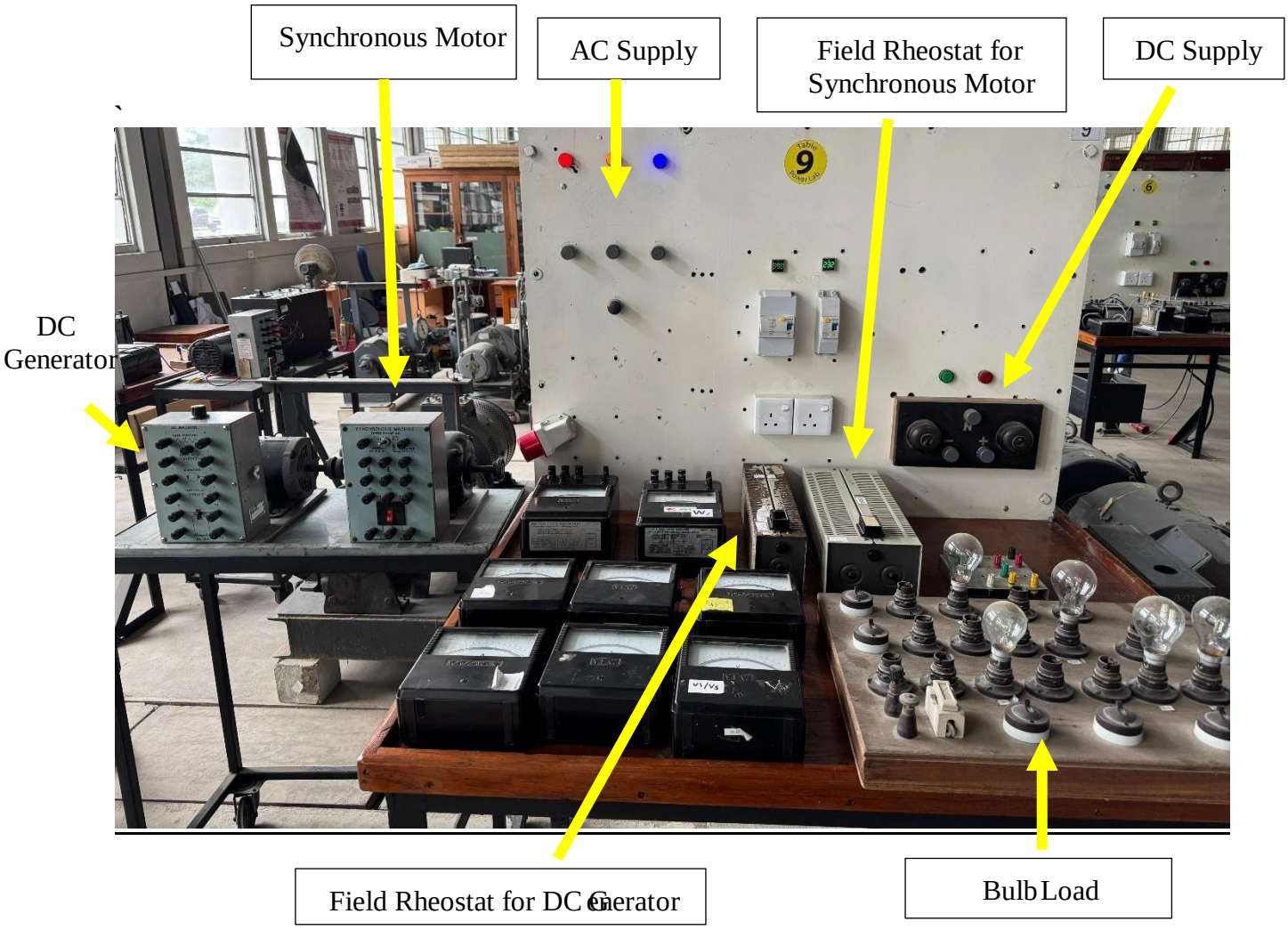


Figure 01: Arrangement of the Setup

Figure 2: Interface and Name Plate of the Synchronous Generator

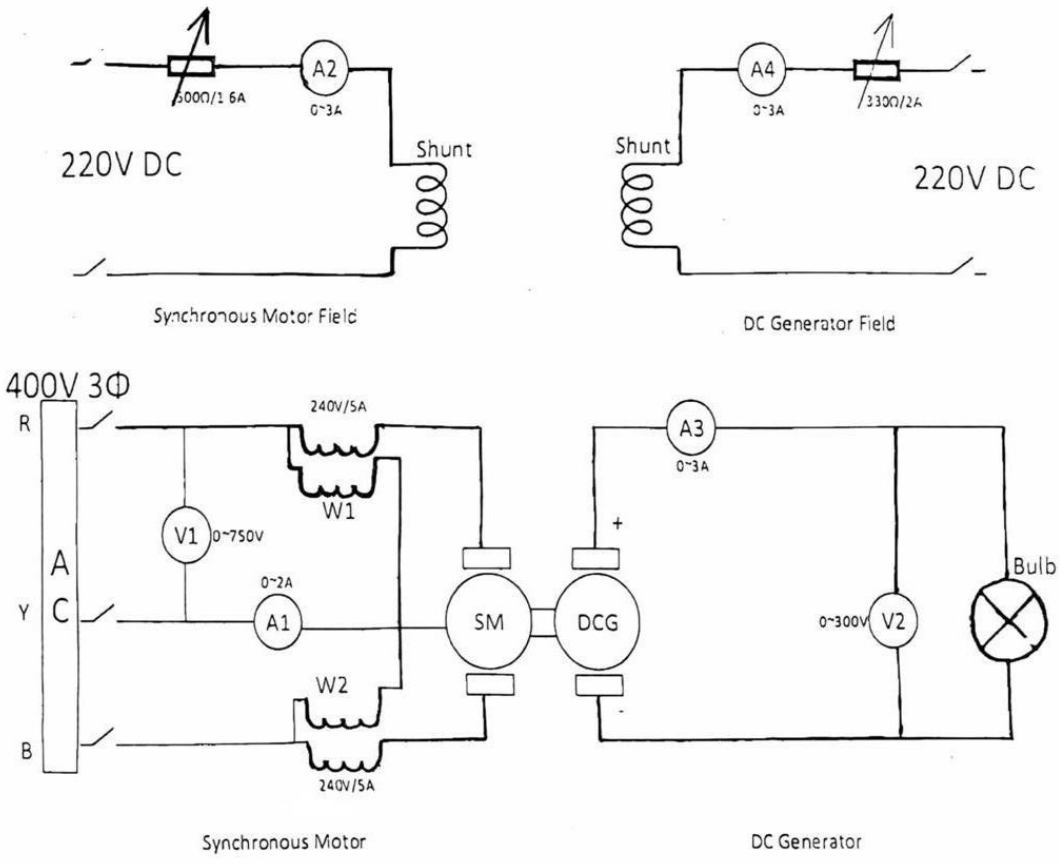


Figure 3: Circuit Diagram

OBSERVATIONS

Motor Field Current ($I_{F,SM}$)	Meter	No Load	Bulb Load1 (60 W)	Bulb Load 2 (160 W)	Bulb Load 3 (260 W)
0.4A	DC Generator Terminal Voltage V2/V	200	194	186	176
	DC Generator Load Current A3/A	0	0.24	0.60	0.94
	Motor Power W1/W	0	24	58	80
	Motor Power W2/W	90	110	145	190
	Motor Line Current A1/A	0.4	0.45	0.58	0.74
0.5A	DC Generator Terminal Voltage V2/V	200	194	186	178
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	20	42	82	110
	Motor Power W2/W	70	90	130	160
	Motor Line Current A1/A	0.32	0.38	0.48	0.64
0.6A	DC Generator Terminal Voltage V2/V	200	196	188	178
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	26	58	96	128
	Motor Power W2/W	60	75	105	140
	Motor Line Current A1/A	0.24	0.30	0.40	0.54
0.7A	DC Generator Terminal Voltage V2/V	200	194	186	178
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	40	70	110	144
	Motor Power W2/W	40	60	90	125
	Motor Line Current A1/A	0.20	0.24	0.34	0.46

Motor Field Current ($I_{F,SM}$)	Meter	No Load	Bulb Load1 (60 W)	Bulb Load 2 (160 W)	Bulb Load 3 (260 W)
0.8A	DC Generator Terminal Voltage V2/V	200	196	188	178
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	54	86	130	164
	Motor Power W2/W	30	50	75	110
	Motor Line Current A1/A	0.20	0.22	0.30	0.42
0.9A	DC Generator Terminal Voltage V2/V	200	194	186	176
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	70	100	142	178
	Motor Power W2/W	150	35	65	80
	Motor Line Current A1/A	0.20	0.24	0.30	0.38
1.0A	DC Generator Terminal Voltage V2/V	200	194	186	176
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	88	118	160	196
	Motor Power W2/W	0	20	50	80
	Motor Line Current A1/A	0.24	0.26	0.32	0.38
1.1A	DC Generator Terminal Voltage V2/V	200	192	184	174
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	108	138	180	214
	Motor Power W2/W	0	5	35	60
	Motor Line Current A1/A	0.32	0.32	0.36	0.42
1.2A	DC Generator Terminal Voltage V2/V	200	196	186	176
	DC Generator Load Current A3/A	0	0.24	0.6	0.94
	Motor Power W1/W	124	158	200	234
	Motor Power W2/W	0	0	20	50
	Motor Line Current A1/A	0.40	0.42	0.44	0.48

CALCULATIONS

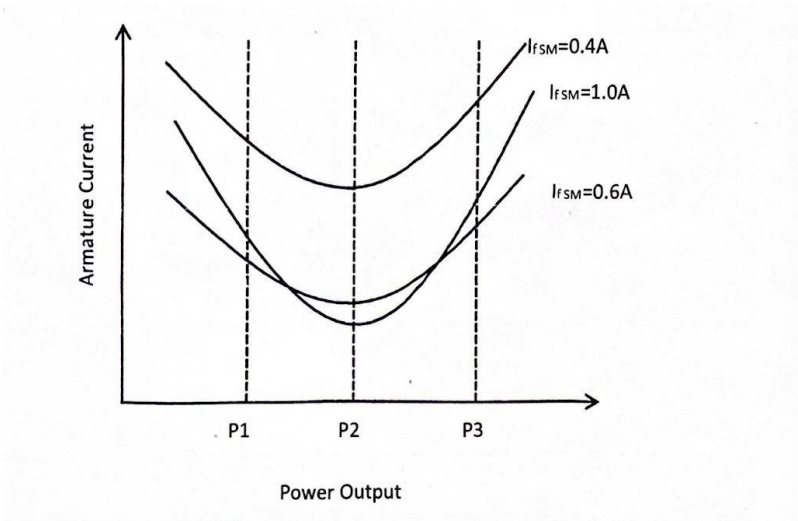
- Calculate the input power, power factor, output power & the efficiency of the synchronous motor for all loads for nine excitation currents.

Note: You should do at least one set (for all no load and 3 bulb loads for any motor field current) of specimen calculation & tabulate values for other eight sets.

TABULATIONS

Tabulate and plot the variations of;

- V curves: Armature current of the synchronous motor Vs Excitation current for constant power output of motor.
- Armature current of the synchronous motor Vs Power output of motor for constant excitation current.



- Power factor of motor Vs Power output of motor for constant excitation current.
- Power factor of motor Vs Excitation for constant power output of motor.
- Efficiency Vs Input power of the synchronous motor for constant excitation current.
  Note: Attach five separate graph sheets.

DISCUSSION

1. Explain different methods can be used to start a synchronous motor.
2. Explain how a synchronous motor can be used for power factor correction.

SPECIMEN CALCULATIONS

For $I_{F,SM} = 0.6$ A

1) Considering set of readings for No load case from Table 1.

$$\text{Input power to the motor} = W_1 + W_2$$
$$= (26 + 60)\ \text{W}$$
$$= 86\ \text{W}$$

$$\text{Power factor of the motor} = \frac{\text{Active power}}{\text{Apparent power}}$$
$$= \frac{W_1 + W_2}{\sqrt{3} \times V_L \times I_L}$$
$$= \frac{86\ \text{W}}{\sqrt{3} \times 400 \times A_1}$$
$$= \frac{86\ \text{W}}{\sqrt{3} \times 400 \times 0.24}$$
$$= 0.517$$

$$\text{Armature resistance loss of DC motor (1)} = {A_3}^2 \times R_a$$
$$= 0 \times 15\ \Omega$$
$$\underline{\underline{= 0\ \text{W}}}$$

$$\text{Output power of DC motor (2)} = V_a \times I_a$$
$$= V_2 \times A_3$$
$$= 200\ \text{V} \times 0$$
$$\underline{\underline{= 0\ \text{W}}}$$

$$\text{Output power of Synchronous motor} = (1) + (2) + \text{Rotational loss}$$
$$= 0 + 0 + 30\ \text{W}$$
$$\underline{\underline{= 30\ \text{W}}}$$

$$\text{Efficiency of Synchronous motor} = \frac{P_{out}}{P_{in}} \times 100\%$$
$$= \frac{30\ \text{W}}{86\ \text{W}} \times 100\%$$
$$= 34.88\ \%$$

For $I_{F,SM}$ = 0.6 A

2) Considering set of readings for Bulb load 1 from Table 1.

$$\text{Input power to the motor} = W_1 + W_2$$
$$= (58 + 75)\ \text{W}$$
$$= 133\ \text{W}$$

$$\text{Power factor of the motor} = \frac{\text{Active power}}{\text{Apparent power}}$$
$$= \frac{W_1 + W_2}{\sqrt{3} \times V_L \times I_L}$$
$$= \frac{133\ \text{W}}{\sqrt{3} \times 400 \times A_1}$$
$$= \frac{133\ \text{W}}{\sqrt{3} \times 400 \times 0.30}$$
$$= 0.640$$

$$\text{Armature resistance loss of DC motor (1)} = {A_3}^2 \times R_a$$
$$= 0.24 \times 0.24 \times 15\ \Omega$$
$$= 0.864\ \text{W}$$

$$\text{Output power of DC motor (2)} = V_a \times I_a$$
$$= V_2 \times A_3$$
$$= 196\ \text{V} \times 0.24\text{A}$$
$$= 47.04\text{W}$$

$$\text{Output power of Synchronous motor} = (1) + (2) + \text{Rotational loss}$$
$$= 0.864 + 47.04 + 30\ \text{W}$$
$$\underline{\underline{= 77.904\ \text{W}}}$$

$$\text{Efficiency of Synchronous motor} = \frac{P_{out}}{P_{in}} \times 100\%$$
$$= \frac{77.904\ \text{W}}{133\ \text{W}} \times 100\%$$
$$\underline{\underline{= 58.57\ \%}}$$

For $I_{F,SM} = 0.6$ A

3) Considering set of readings for Bulb load 2 from Table 1.

$$\text{Input power to the motor} = W_1 + W_2$$
$$= (96 + 105)\ \text{W}$$
$$= 201\ \text{W}$$

$$\text{Power factor of the motor} = \frac{\text{Active power}}{\text{Apparent power}}$$
$$= \frac{W_1 + W_2}{\sqrt{3} \times V_L \times I_L}$$
$$= \frac{201\text{W}}{\sqrt{3} \times 400 \times A_1}$$
$$= \frac{201\ \text{W}}{\sqrt{3} \times 400 \times 0.40}$$
$$= 0.725$$

$$\text{Armature resistance loss of DC motor (1)} = {A_3}^2 \times R_a$$
$$= 0.6 \times 0.6 \times 15\ \Omega$$
$$\underline{\underline{= 5.4\ \text{W}}}$$

$$\text{Output power of DC motor (2)} = V_a \times I_a$$
$$= V_2 \times A_3$$
$$= 188\ \text{V} \times 0.6\text{A}$$
$$= 112.8\ \text{W}$$

$$\text{Output power of Synchronous motor} = (1) + (2) + \text{Rotational loss}$$
$$= 5.4 + 112.8 + 30\ \text{W}$$
$$= 148.2\ \text{W}$$

$$\text{Efficiency of Synchronous motor} = \frac{P_{out}}{P_{in}} \times 100\%$$
$$= \frac{148.2\ \text{W}}{201\ \text{W}} \times 100\%$$
$$= 73.73\ \%$$

For $I_{F,SM} = 0.6$ A

4) Considering set of readings for Bulb load 3 from Table 1.

$$\text{Input power to the motor} = W_1 + W_2$$
$$= (128 + 140)\ \text{W}$$
$$= 268\ \text{W}$$

$$\text{Power factor of the motor} = \frac{\text{Active power}}{\text{Apparent power}}$$
$$= \frac{W_1 + W_2}{\sqrt{3} \times V_L \times I_L}$$
$$= \frac{268\ \text{W}}{\sqrt{3} \times 400 \times A_1}$$
$$= \frac{268\ \text{W}}{\sqrt{3} \times 400 \times 0.54}$$
$$= 0.716$$

$$\text{Armature resistance loss of DC motor (1)} = {A_3}^2 \times R_a$$
$$= 0.94 \times 0.94 \times 15\ \Omega$$
$$\underline{\underline{= 13.254\ \text{W}}}$$

$$\text{Output power of DC motor (2)} = V_a \times I_a$$
$$= V_2 \times A_3$$
$$= 178\ \text{V} \times 0.94\text{A}$$
$$= 167.32\ \text{W}$$

$$\text{Output power of Synchronous motor} = (1) + (2) + \text{Rotational loss}$$
$$= 13.254 + 167.32 + 30\ \text{W}$$
$$= 210.574\ \text{W}$$

$$\text{Efficiency of Synchronous motor} = \frac{P_{out}}{P_{in}} \times 100\%$$
$$= \frac{210.574\ \text{W}}{268\ \text{W}} \times 100\%$$
$$= 78.57\ \%$$

RESULTS

Table 02: The input power, power factor, output power and efficiency with excitation currents

Motor Field Current ($I_{F,SM}$)	Meter	No Load	Bulb Load1 (60 W)	Bulb Load 2 (160 W)	Bulb Load 3 (260 W)
0.4A	Input Power	90	134	203	270
	Output Power	30	77.42	147	208.69
	Efficiency	33.33	57.78	72.41	77.29
	Power Factor	0.324	0.430	0.505	0.527
0.5A	Input Power	90	132	212	270
	Output Power	30	77.42	147	210.57
	Efficiency	33.33	58.65	69.34	77.99
	Power Factor	0.406	0.50138	0.63749	0.608
0.6A	Input Power	86	133	201	268
	Output Power	30	77.90	148.20	210.57
	Efficiency	34.88	58.57	73.73	78.57
	Power Factor	0.517	0.640	0.725	0.716
0.7A	Input Power	80	130	200	269
	Output Power	30	77.42	147	210.57
	Efficiency	37.50	59.56	73.5	78.28
	Power Factor	0.577	0.782	0.849	0.844
0.8A	Input Power	84	136	205	274
	Output Power	30	77.90	148.2	210.57
	Efficiency	35.71	57.28	72.29	76.85
	Power Factor	0.606	0.892	0.986	0.942
0.9A	Input Power	85	135	207	258
	Output Power	30	77.424	147	208.694
	Efficiency	35.29	57.35	71.01	80.89
	Power Factor	0.613	0.812	0.996	0.980
1.0A	Input Power	88	138	210	276
	Output Power	30	77.424	147	208.69
	Efficiency	34.09	56.10	70	75.61
	Power Factor	0.529	0.766	0.947	1.048
1.1A	Input Power	108	143	215	274
	Output Power	30	76.944	145.8	206.81
	Efficiency	27.78	53.81	67.81	75.48
	Power Factor	0.487	0.645	0.862	0.942
1.2A	Input Power	124	158	220	284
	Output Power	30	77.904	147	208.69
	Efficiency	24.19	49.30	66.82	73.484
	Power Factor	0.447	0.542	0.721	0.854

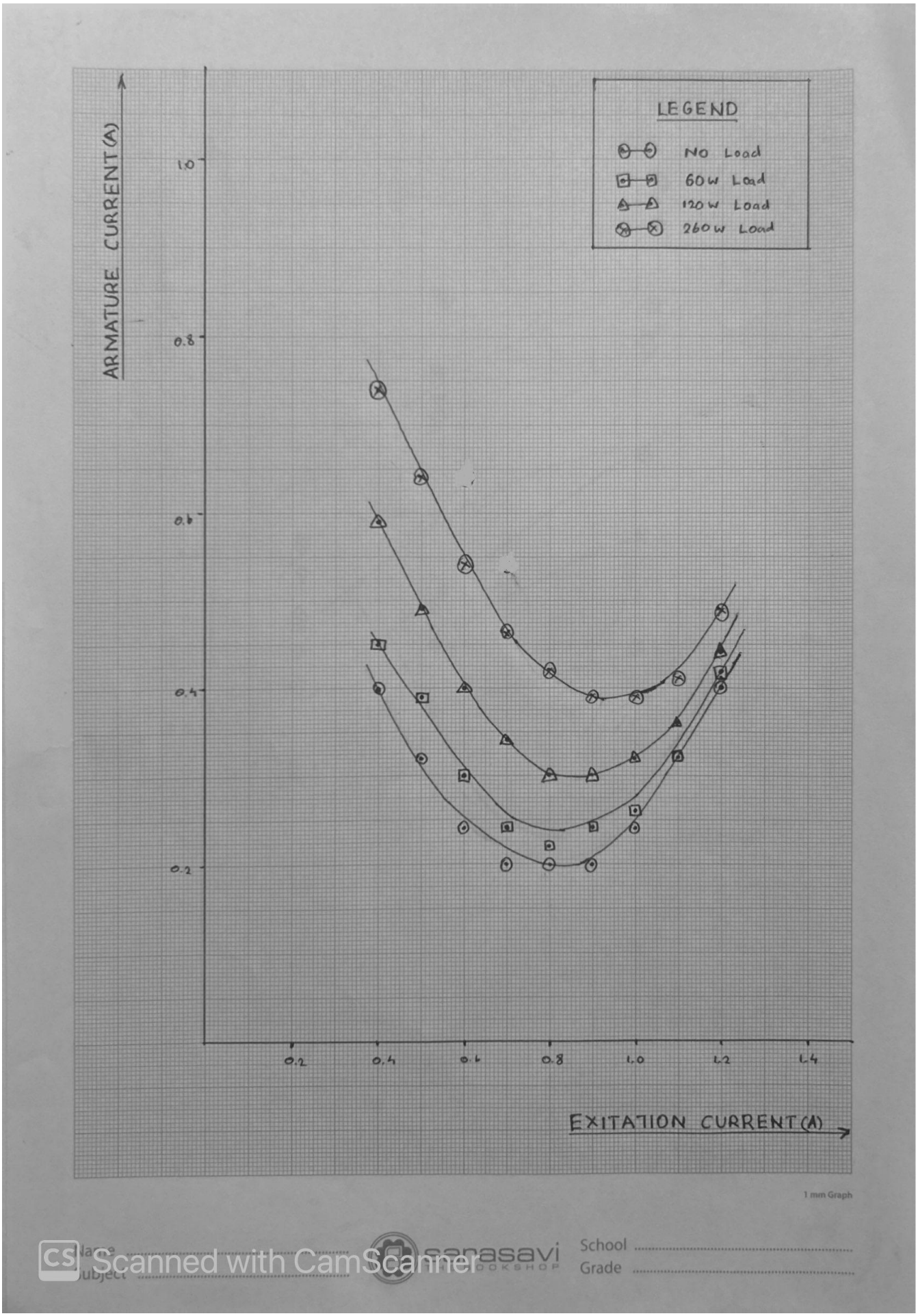


Figure 04 : Variation of armature current of the synchronous motor and Excitation current for constant power output of motor.

Table 03: Armature current of the synchronous motor Vs Excitation current for constant power output of motor.

Excitation Current (IF) (A)	No Load (A)	Bulb Load 1 (60W) (A)	Bulb Load 2 (160W) (A)	Bulb Load 3 (260W) (A)
0.4	0.4	0.45	0.58	0.74
0.5	0.32	0.38	0.48	0.64
0.6	0.24	0.30	0.40	0.54
0.7	0.20	0.24	0.34	0.46
0.8	0.20	0.22	0.30	0.42
0.9	0.20	0.24	0.30	0.38
1.0	0.24	0.26	0.32	0.38
1.1	0.32	0.32	0.36	0.42
1.2	0.40	0.42	0.44	0.48

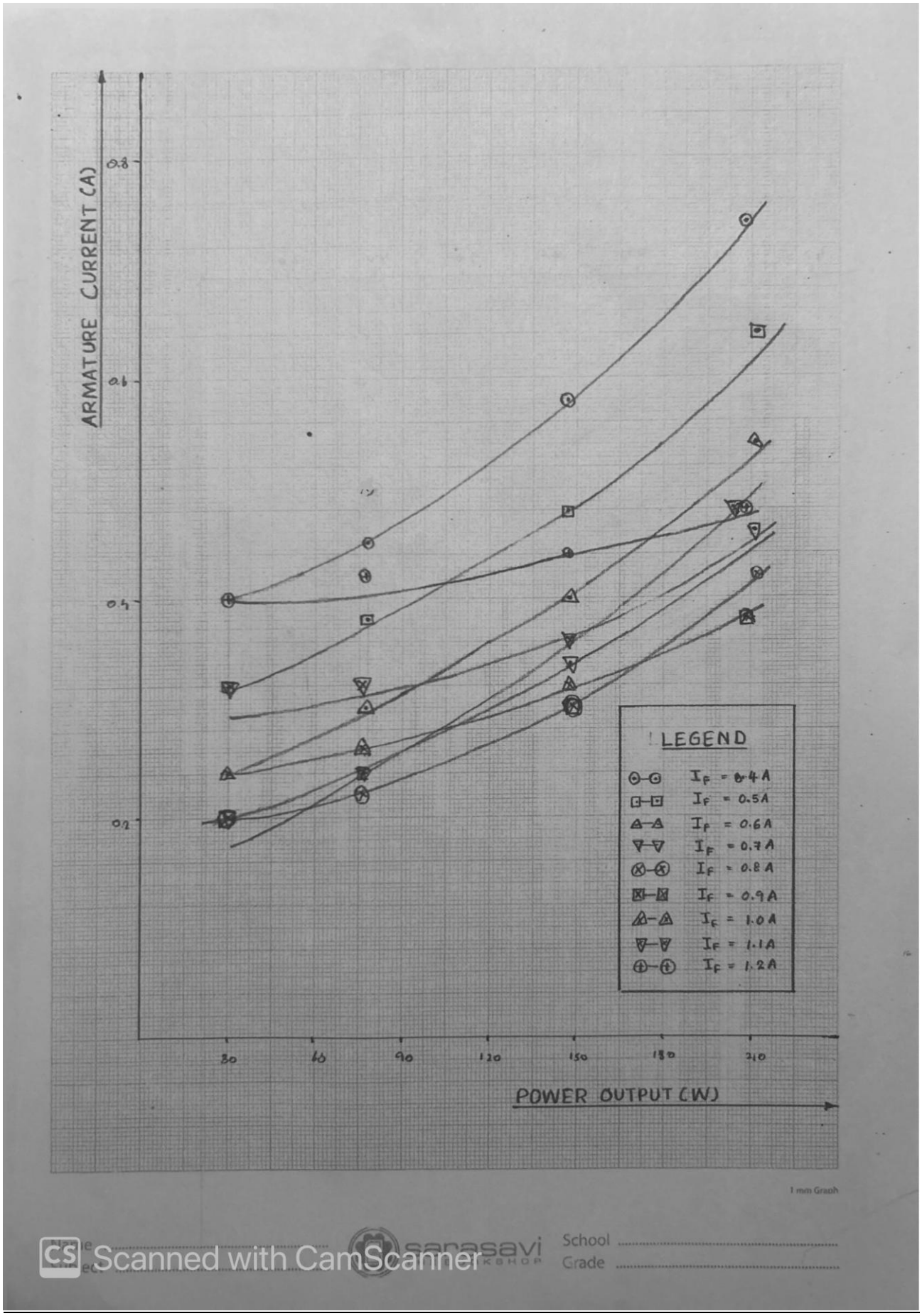


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Figure 05 : Variation of armature current of the synchronous motor and Power output of motor for constant excitation current

Table 04: Armature current of the synchronous motor Vs Power output of motor for constant excitation current.

Field Current (A)	Load Type	Armature Current (A)	Output Power (W)
0.4	No Load	0.40	30.000
	Bulb Load 1	0.45	77.420
	Bulb Load 2	0.58	147.000
	Bulb Load 3	0.74	208.690
0.5	No Load	0.32	30.000
	Bulb Load 1	0.38	77.420
	Bulb Load 2	0.48	147.000
	Bulb Load 3	0.64	210.570
0.6	No Load	0.24	30.000
	Bulb Load 1	0.30	77.900
	Bulb Load 2	0.40	148.200
	Bulb Load 3	0.54	210.570
0.7	No Load	0.20	30.000
	Bulb Load 1	0.24	77.420
	Bulb Load 2	0.34	147.000
	Bulb Load 3	0.46	210.570
0.8	No Load	0.20	30.000
	Bulb Load 1	0.22	77.900
	Bulb Load 2	0.30	148.200
	Bulb Load 3	0.42	210.570
0.9	No Load	0.20	30.000
	Bulb Load 1	0.24	77.424
	Bulb Load 2	0.30	147.000
	Bulb Load 3	0.38	208.694
1.0	No Load	0.24	30.000
	Bulb Load 1	0.26	77.424
	Bulb Load 2	0.32	147.000
	Bulb Load 3	0.38	208.690
1.1	No Load	0.32	30.000
	Bulb Load 1	0.32	76.944
	Bulb Load 2	0.36	145.800
	Bulb Load 3	0.48	206.810
1.2	No Load	0.40	30.000
	Bulb Load 1	0.42	77.904
	Bulb Load 2	0.44	147.000
	Bulb Load 3	0.48	208.690

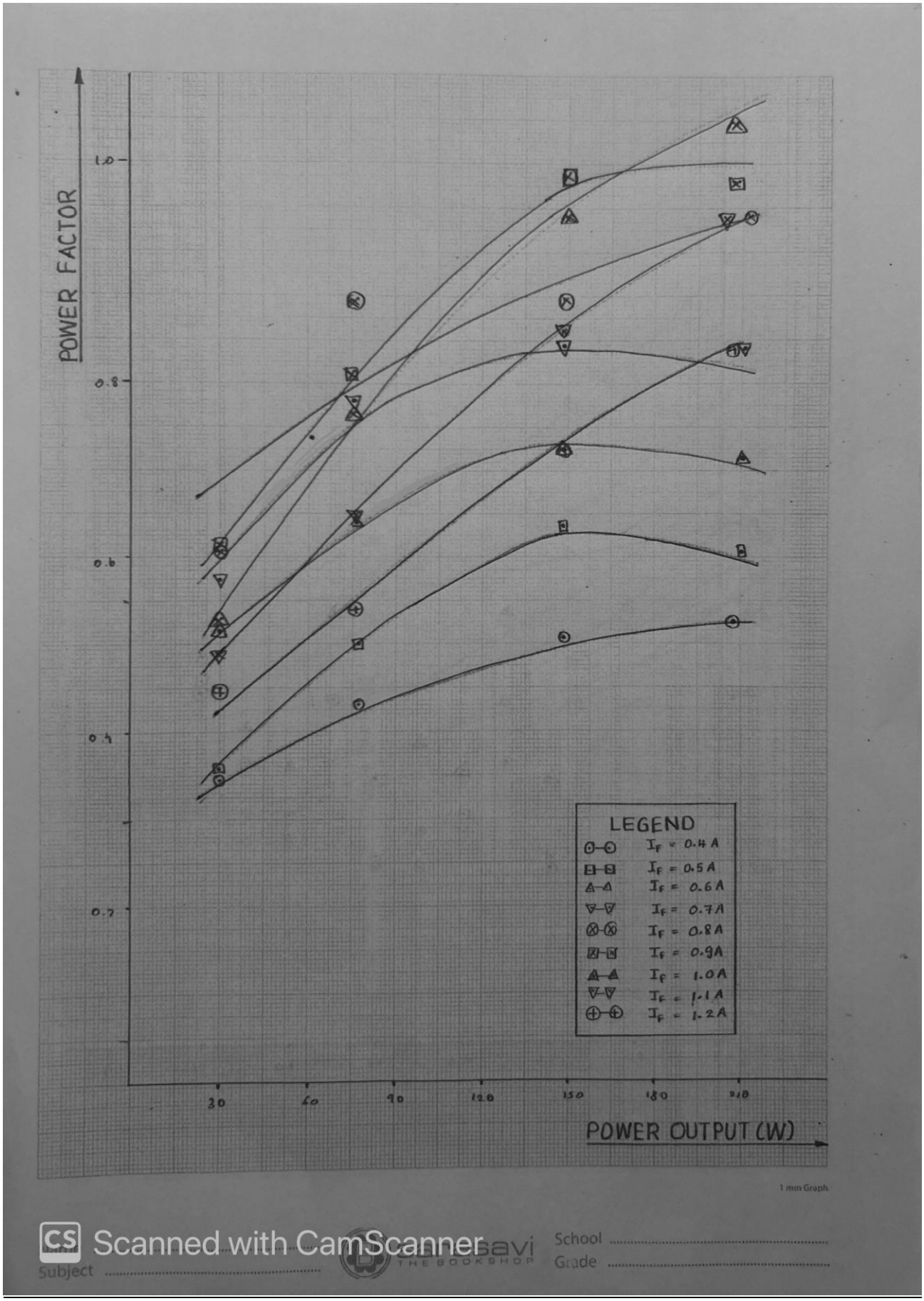

Figure 06 : Variation of power factor of motor and power output of motor for constant excitation current

Table 05: Power factor of motor Vs Power output of motor for constant excitation current.

Field Current (A)	Load Type	Power Factor	Output Power (W)
0.4	No Load	0.324	30.000
	Bulb Load 1	0.430	77.420
	Bulb Load 2	0.505	147.000
	Bulb Load 3	0.527	208.690
0.5	No Load	0.406	30.000
	Bulb Load 1	0.501	77.420
	Bulb Load 2	0.637	147.000
	Bulb Load 3	0.608	210.570
0.6	No Load	0.517	30.000
	Bulb Load 1	0.640	77.900
	Bulb Load 2	0.725	148.200
	Bulb Load 3	0.716	210.570
0.7	No Load	0.577	30.000
	Bulb Load 1	0.782	77.420
	Bulb Load 2	0.849	147.000
	Bulb Load 3	0.844	210.570
0.8	No Load	0.606	30.000
	Bulb Load 1	0.892	77.900
	Bulb Load 2	0.986	148.200
	Bulb Load 3	0.942	210.570
0.9	No Load	0.613	30.000
	Bulb Load 1	0.812	77.424
	Bulb Load 2	0.996	147.000
	Bulb Load 3	0.980	208.694
1.0	No Load	0.529	30.000
	Bulb Load 1	0.766	77.424
	Bulb Load 2	0.947	147.000
	Bulb Load 3	1.048	208.690
1.1	No Load	0.487	30.000
	Bulb Load 1	0.645	76.944
	Bulb Load 2	0.862	145.800
	Bulb Load 3	0.942	206.810
1.2	No Load	0.447	30.000
	Bulb Load 1	0.542	77.904
	Bulb Load 2	0.721	147.000
	Bulb Load 3	0.854	208.690

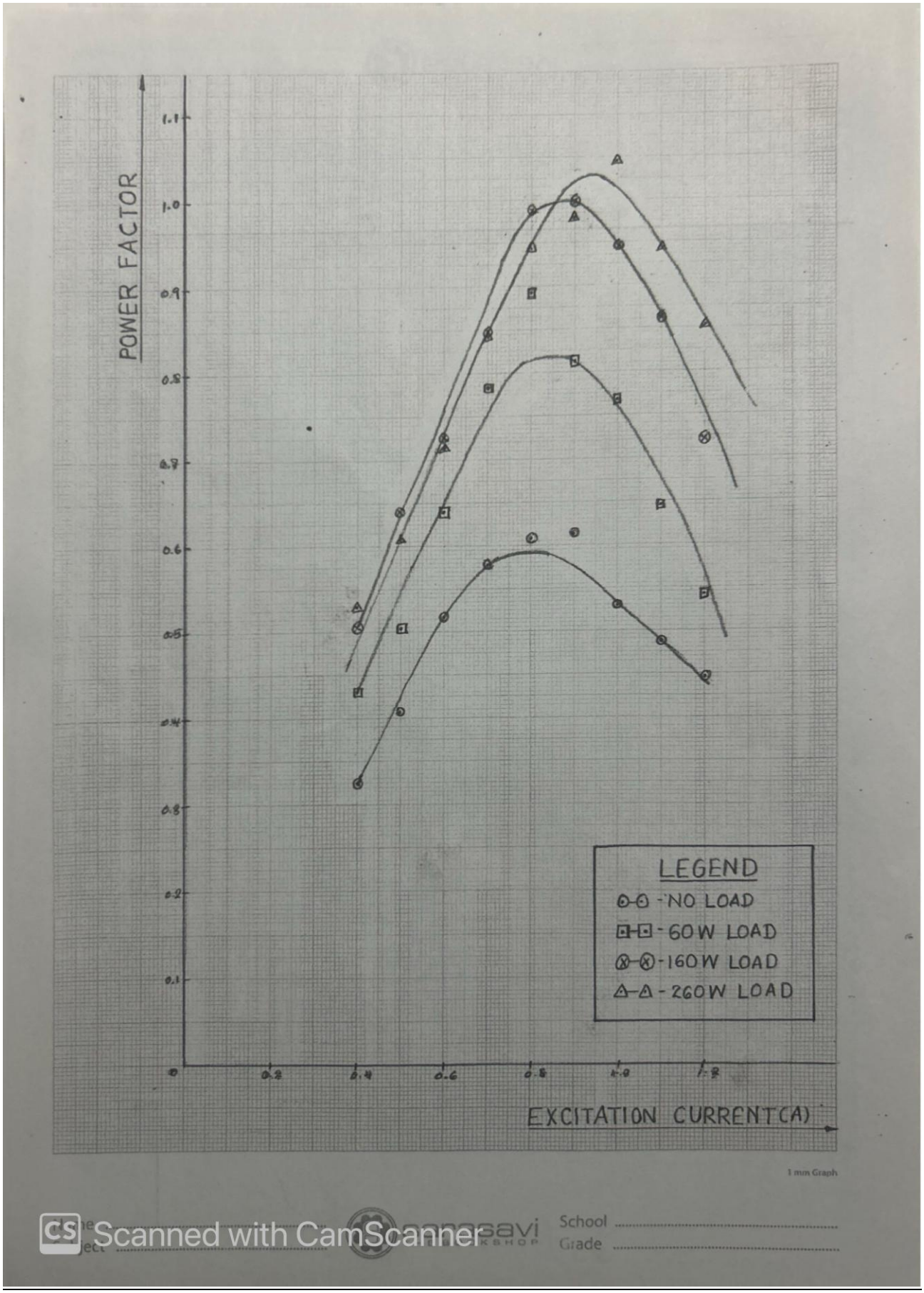

Figure 06 : Variation of power factor of motor and power output of motor for constant excitation current.

Table 06:Power factor of motor Vs Excitation for constant power output of motor.

Excitation Current (IF)	PF at No Load	PF at Bulb Load 1 (60W)	PF at Bulb Load 2 (160W)	PF at Bulb Load 3 (260W)
0.4	0.324	0.430	0.505	0.527
0.5	0.406	0.501	0.637	0.608
0.6	0.517	0.640	0.725	0.716
0.7	0.577	0.782	0.849	0.844
0.8	0.606	0.892	0.986	0.942
0.9	0.613	0.812	0.996	0.980
1.0	0.529	0.766	0.947	1.048
1.1	0.487	0.645	0.862	0.942
1.2	0.447	0.542	0.721	0.854

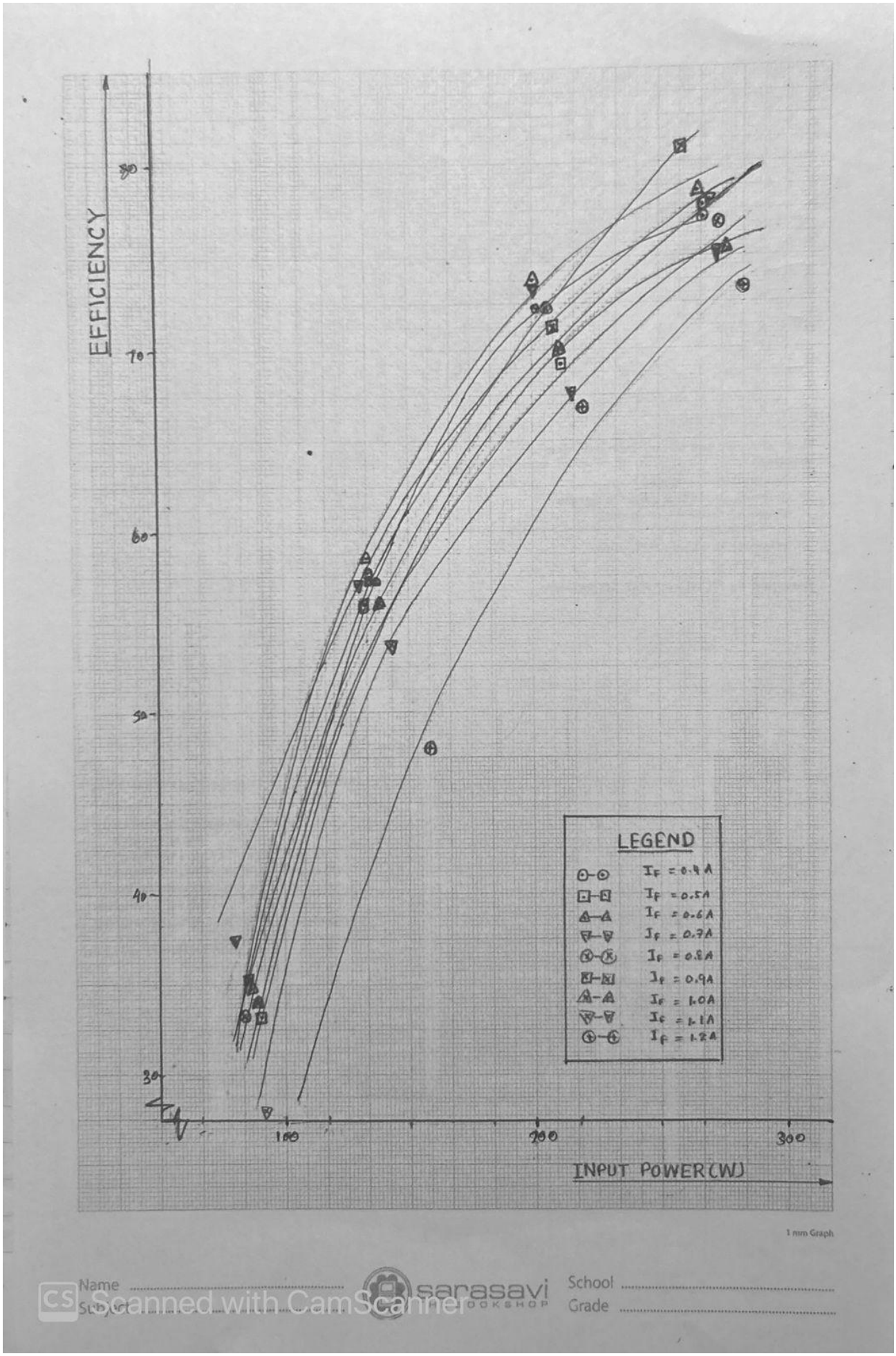

Figure 06 : Variation of power factor of motor and power output of motor for constant excitation current.

Table 07: Efficiency Vs Input power of the synchronous motor for constant excitation current.

Field Current (A)	Load Type	Efficiency (%)	Input Power (W)
0.4	No Load	33.33	90
	Bulb Load 1	57.78	134
	Bulb Load 2	72.41	203
	Bulb Load 3	77.29	270
0.5	No Load	33.33	90
	Bulb Load 1	58.65	132
	Bulb Load 2	69.34	212
	Bulb Load 3	77.99	270
0.6	No Load	34.88	86
	Bulb Load 1	58.57	133
	Bulb Load 2	73.73	201
	Bulb Load 3	78.57	268
0.7	No Load	37.50	80
	Bulb Load 1	59.56	130
	Bulb Load 2	73.50	200
	Bulb Load 3	78.28	269
0.8	No Load	35.71	84
	Bulb Load 1	57.28	136
	Bulb Load 2	72.29	205
	Bulb Load 3	76.85	274
0.9	No Load	35.29	85
	Bulb Load 1	57.35	135
	Bulb Load 2	71.01	207
	Bulb Load 3	80.89	258
1.0	No Load	34.09	88
	Bulb Load 1	56.10	138
	Bulb Load 2	70.00	210
	Bulb Load 3	75.61	276
1.1	No Load	27.78	108
	Bulb Load 1	53.81	143
	Bulb Load 2	67.81	215
	Bulb Load 3	75.48	274
1.2	No Load	24.19	124
	Bulb Load 1	49.30	158
	Bulb Load 2	66.82	220
	Bulb Load 3	73.48	284

DISCUSSION

1. Explain different methods can be used to start a synchronous motor.

A synchronous motor is not self-starting by nature. When a three-phase AC supply is applied to the stator, it produces a magnetic field that rotates at synchronous speed. However, because the rotor has mechanical inertia, it cannot instantly accelerate and match this high speed. As a result, there is no continuous average starting torque, and the motor fails to start on its own. Therefore, special starting methods are required to bring the rotor close to synchronous speed before normal operation begins.

One common method used is the damper winding method. In this technique, heavy copper or brass bars are embedded in the rotor pole faces and short-circuited at both ends with end rings, similar to the construction of a squirrel-cage induction motor. When the stator is energized, the rotating magnetic field induces currents in these damper bars. These induced currents produce torque, allowing the motor to start and accelerate like an induction motor. Once the rotor reaches approximately 95% of synchronous speed, DC excitation is applied to the field winding. At this point, the rotor poles lock with the rotating magnetic field of the stator, and the motor runs steadily at synchronous speed.

Another method is auxiliary motor starting, also known as the pony motor method. In this approach, a small external motor is mechanically coupled to the shaft of the synchronous motor. This auxiliary motor drives the rotor up to synchronous speed before the main motor is connected to the supply. After synchronization with the AC grid and application of DC excitation to the rotor, the auxiliary motor is disconnected. In some cases, if a DC motor is used as the pony motor, it can later function as the DC exciter for the synchronous motor.

A more modern approach is starting with a Variable Frequency Drive (VFD). Instead of forcing the rotor to catch up with a rapidly rotating magnetic field, this method initially supplies the stator with a very low frequency. Since synchronous speed is directly proportional to supply frequency, the magnetic field rotates slowly at first, allowing the rotor to lock in easily. The VFD then gradually increases both frequency and voltage until the motor reaches its rated operating speed. This method provides smooth starting, better control, and reduced mechanical stress compared to traditional techniques.

2. Explain how a synchronous motor can be used for power factor correction.

One of the most important and unique features of a synchronous motor is its ability to control power factor simply by adjusting the DC excitation supplied to its rotor. This makes it extremely valuable in industrial power systems. When a synchronous motor is operated without any mechanical load purely for improving power factor, it is called a synchronous condenser or synchronous phase modifier.

The key idea behind this operation lies in the DC field current. The amount of DC excitation determines the strength of the magnetic flux produced by the rotor. The stator then draws current from the AC supply depending on whether this rotor flux is insufficient, adequate, or excessive. In simple terms, the stator current automatically adjusts itself to maintain the required magnetic balance in the system.

If the motor is under-excited, meaning the DC field current is too low, the rotor produces less magnetic flux than required. To compensate for this shortage, the stator draws lagging reactive power (VARs) from the supply. In this condition, the motor behaves like an inductor and operates at a lagging power factor.

When the motor is normally excited, the DC field current produces exactly the required magnetic flux. In this case, the stator only draws active power to overcome mechanical losses such as friction and windage. The reactive power exchange becomes nearly zero, and the motor operates at unity power factor.

If the motor is over-excited, the DC field current is higher than necessary, and the rotor produces excess magnetic flux. To balance this excess, the stator sends leading reactive power back into the supply. Now the motor behaves like a variable capacitor and operates at a leading power factor.

This characteristic is particularly useful in industrial environments. Most industrial loads, such as induction motors, transformers, and fluorescent lighting systems, require magnetic fields for operation. Because of this, they draw lagging reactive power from the supply, which reduces the overall power factor of the facility. A low power factor increases line current, causes higher I^2R losses in transmission lines, and can even result in penalty charges from utility companies.

To solve this problem, an over-excited synchronous motor is connected in parallel with the plant's main power bus. Acting as a synchronous condenser, it supplies leading reactive power that cancels out the lagging reactive power drawn by inductive loads. Since the excitation can be adjusted using a rheostat or an automatic voltage regulator (AVR), the amount of reactive power supplied can be controlled smoothly and continuously. As plant loads vary throughout the day, the synchronous condenser can dynamically maintain the overall power factor close to unity, improving efficiency and reducing operational costs.

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